ใบงานที่ 1

เรื่อง การทดลองวงจรมัลติเพลกเซอร์และวงจรดีมัลติเพลกเซอร์

(Multiplexer and Demultiplexer)

จัดทำโดย

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รหัส ENGCE200_SEC1

เสนอ

อาจารย์สมนึก สุระธง

ใบงานนี้เป็นส่วนหนึ่งของวิชา

ENGCE200

การออกแบบดิจิทัล

(Digital Systems Design)

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มหาวิทยาลัยเทคโนโลยีราชมงคลล้านนา ภาคพายัพ เชียงใหม่

ภาคเรียนที่ 1 ปีการศึกษา 2568

ใบงานที่ 1

ชื่อบทเรียน การทดลองวงจรมัลติเพลกเซอร์และวงจรดีมัลติเพลกเซอร์

(Multiplexer and Demultiplexer)

จุดประสงค์การสอน

ปฏิบัติการทดลองสร้างวงจรคอมไบเนชันด้วยลอจิกเกตพื้นฐาน

ออกแบบและทดสอบวงจรมัลติเพล็กซ์

ออกแบบและทดสอบวงจรดีมัลติเพล็กซ์

1. ทฤษฎี

วิธีการออกแบบวงจรมัลติเพล็กซ์ (Multiplexer)

วิธีการออกแบบวงจรดีมัลติเพล็กซ์ (Demultiplexer)

2. จงออกแบบวงจรต่อไปนี้

1. วงจรมัลติเพล็กซ์ (16 : 1 Multiplexer)

2. วงจรดีมัลติเพล็กซ์ (1 : 16 Demultiplexer)

3. ผลการทดลอง

ทดลองโดยการนำ Output Multiplexer ต่อ เข้า Input Demultiplexer แล้วเลือก select อิสระ

4. สรุปผลการทดลอง

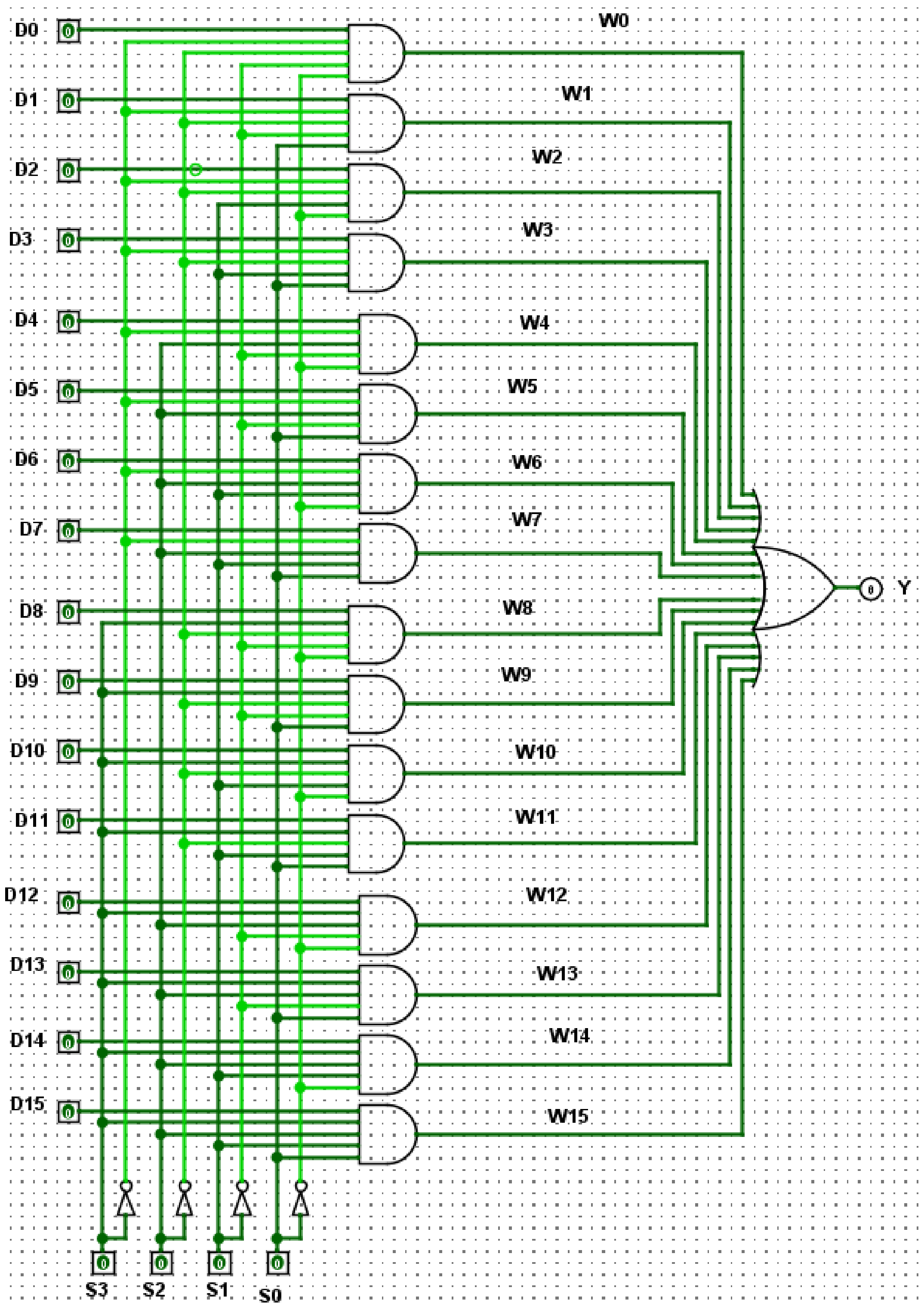
5. คำถามหลังการทดลอง

1. บอกหลักการทำงานวงจรมัลติเพล็กซ์

2. บอกหลักการทำงานวงจรดีมัลติเพล็กซ์

3. บอกวิธีการเชื่อมต่อทางไกลโดยสายส่งคู่เดียว

Multiplexer 16:1 logic



MUX16to1.vhdl

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity MUX16to1 is
  port (
    D : in std_logic_vector(15 downto 0);
    S : in std_logic_vector(3 downto 0);
    Y : out std_logic
  );
end MUX16to1;

architecture GateLevel of MUX16to1 is

  signal W : std_logic_vector(15 downto 0);

begin

  W(0) <= D(0) and (not S(3)) and (not S(2)) and (not S(1)) and (not
S(0));
  W(1) <= D(1) and (not S(3)) and (not S(2)) and (not S(1)) and S(0);
  W(2) <= D(2) and (not S(3)) and (not S(2)) and S(1) and (not S(0));
  W(3) <= D(3) and (not S(3)) and (not S(2)) and S(1) and S(0);

  W(4) <= D(4) and (not S(3)) and S(2) and (not S(1)) and (not S(0));
  W(5) <= D(5) and (not S(3)) and S(2) and (not S(1)) and S(0);
  W(6) <= D(6) and (not S(3)) and S(2) and S(1) and (not S(0));
  W(7) <= D(7) and (not S(3)) and S(2) and S(1) and S(0);

  W(8)  <= D(8) and S(3) and (not S(2)) and (not S(1)) and (not S(0));
  W(9)  <= D(9) and S(3) and (not S(2)) and (not S(1)) and S(0);
```

```
  W(10) <= D(10) and S(3) and (not S(2)) and S(1) and (not S(0));
  W(11) <= D(11) and S(3) and (not S(2)) and S(1) and S(0);

  W(12) <= D(12) and S(3) and S(2) and (not S(1)) and (not S(0));
  W(13) <= D(13) and S(3) and S(2) and (not S(1)) and S(0);
  W(14) <= D(14) and S(3) and S(2) and S(1) and (not S(0));
  W(15) <= D(15) and S(3) and S(2) and S(1) and S(0);

  Y <= W(0) or W(1) or W(2) or W(3) or
    W(4) or W(5) or W(6) or W(7) or
    W(8) or W(9) or W(10) or W(11) or
    W(12) or W(13) or W(14) or W(15);

end GateLevel;
```

tb_MUX16to1.vhdl

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity tb_MUX16to1 is
end tb_MUX16to1;

architecture sim of tb_MUX16to1 is

  signal D0  : std_logic := '0';
  signal D1  : std_logic := '0';
  signal D2  : std_logic := '0';
  signal D3  : std_logic := '0';
  signal D4  : std_logic := '0';
  signal D5  : std_logic := '0';
  signal D6  : std_logic := '0';
  signal D7  : std_logic := '0';
  signal D8  : std_logic := '0';
  signal D9  : std_logic := '0';
  signal D10 : std_logic := '0';
  signal D11 : std_logic := '0';
  signal D12 : std_logic := '0';
  signal D13 : std_logic := '0';
  signal D14 : std_logic := '0';
  signal D15 : std_logic := '0';

  signal D : std_logic_vector(15 downto 0);

  signal S : std_logic_vector(3 downto 0);
  signal Y : std_logic;

begin
```

```
D <= D15 & D14 & D13 & D12 &
  D11 & D10 & D9 & D8 &
  D7 & D6 & D5 & D4 &
  D3 & D2 & D1 & D0;

uut : entity work.MUX16to1
  port map
  (
    D => D,
    S => S,
    Y => Y
  );

process
begin

  D0 <= '1';
  S  <= "0000";
  wait for 10 ns;
  D0 <= '0';

  D1 <= '1';
  S  <= "0001";
  wait for 10 ns;
  D1 <= '0';

  D2 <= '1';
  S  <= "0010";
  wait for 10 ns;
  D2 <= '0';
```

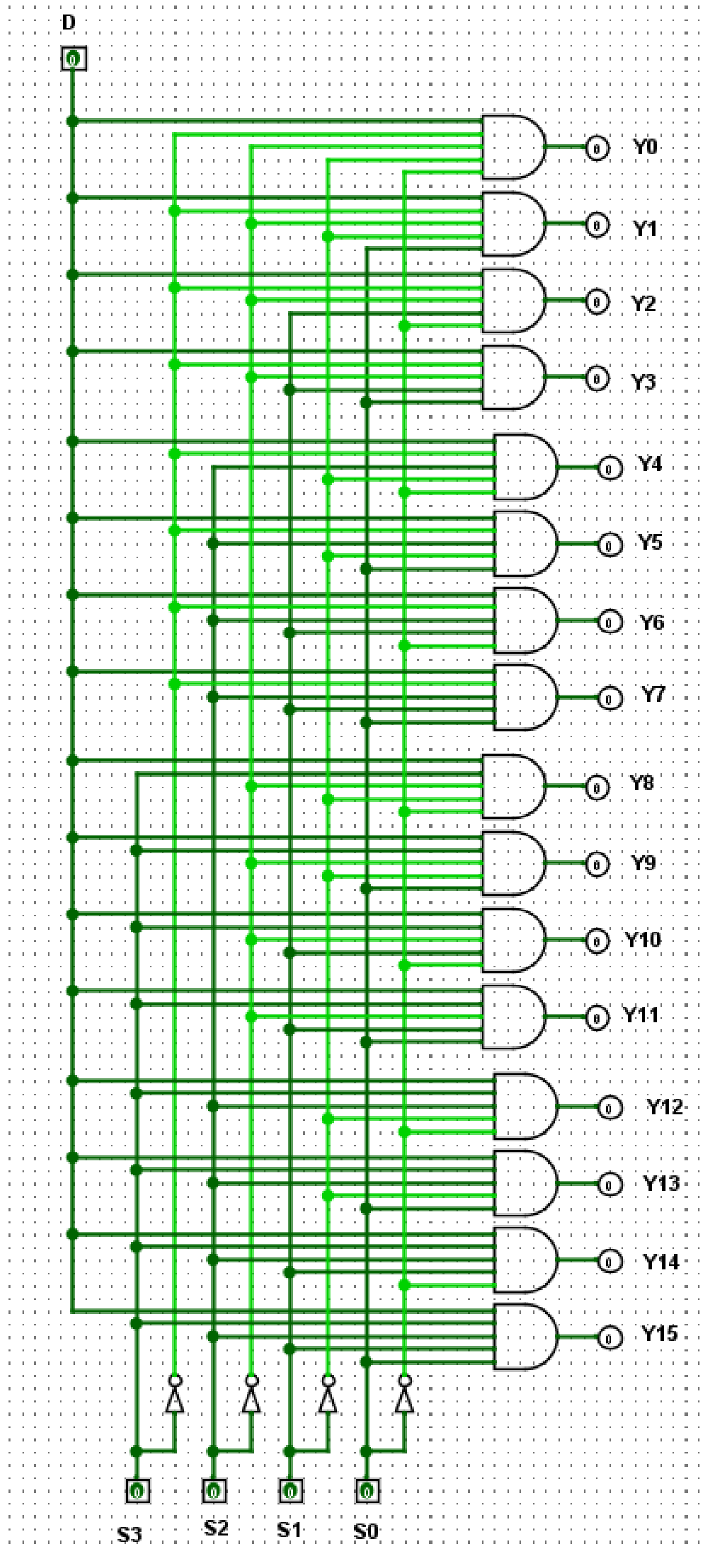
```
D3 <= '1';
S  <= "0011";
wait for 10 ns;
D3 <= '0';

D4 <= '1';
S  <= "0100";
wait for 10 ns;
D4 <= '0';

D5 <= '1';
S  <= "0101";
wait for 10 ns;
D5 <= '0';

D6 <= '1';
S  <= "0110";
wait for 10 ns;
D6 <= '0';

D7 <= '1';
S  <= "0111";
wait for 10 ns;
D7 <= '0';

D8 <= '1';
S  <= "1000";
wait for 10 ns;
D8 <= '0';

D9 <= '1';
S  <= "1001";
wait for 10 ns;
```

```
D9 <= '0';

D10 <= '1';
S   <= "1010";
wait for 10 ns;
D10 <= '0';

D11 <= '1';
S   <= "1011";
wait for 10 ns;
D11 <= '0';

D12 <= '1';
S   <= "1100";
wait for 10 ns;
D12 <= '0';

D13 <= '1';
S   <= "1101";
wait for 10 ns;
D13 <= '0';

D14 <= '1';
S   <= "1110";
wait for 10 ns;
D14 <= '0';

D15 <= '1';
S   <= "1111";
wait for 10 ns;
D15 <= '0';
```

```
    wait;

  end process;

end sim;
```

MUX16-1-wave.vcd

Demultiplexer 1:16 logic



DMUX1to16.vhdl

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity DMUX1to16 is
  port (
    D : in std_logic;
    S : in std_logic_vector(3 downto 0);
    Y : out std_logic_vector(15 downto 0)
  );
end DMUX1to16;

architecture GateLevel of DMUX1to16 is
begin

  Y(0) <= D and (not S(3)) and (not S(2)) and (not S(1)) and (not
S(0)); -- 0000
  Y(1) <= D and (not S(3)) and (not S(2)) and (not S(1)) and S(0);
  Y(2) <= D and (not S(3)) and (not S(2)) and S(1) and (not S(0));
  Y(3) <= D and (not S(3)) and (not S(2)) and S(1) and S(0);

  Y(4) <= D and (not S(3)) and S(2) and (not S(1)) and (not S(0));
  Y(5) <= D and (not S(3)) and S(2) and (not S(1)) and S(0);
  Y(6) <= D and (not S(3)) and S(2) and S(1) and (not S(0));
  Y(7) <= D and (not S(3)) and S(2) and S(1) and S(0);

  Y(8)  <= D and S(3) and (not S(2)) and (not S(1)) and (not S(0));
  Y(9)  <= D and S(3) and (not S(2)) and (not S(1)) and S(0);
  Y(10) <= D and S(3) and (not S(2)) and S(1) and (not S(0));
  Y(11) <= D and S(3) and (not S(2)) and S(1) and S(0);

  Y(12) <= D and S(3) and S(2) and (not S(1)) and (not S(0));
```

```
  Y(13) <= D and S(3) and S(2) and (not S(1)) and S(0);
  Y(14) <= D and S(3) and S(2) and S(1) and (not S(0));
  Y(15) <= D and S(3) and S(2) and S(1) and S(0); -- 1111

end GateLevel;
```

tb_DMUX1to16.vhdl

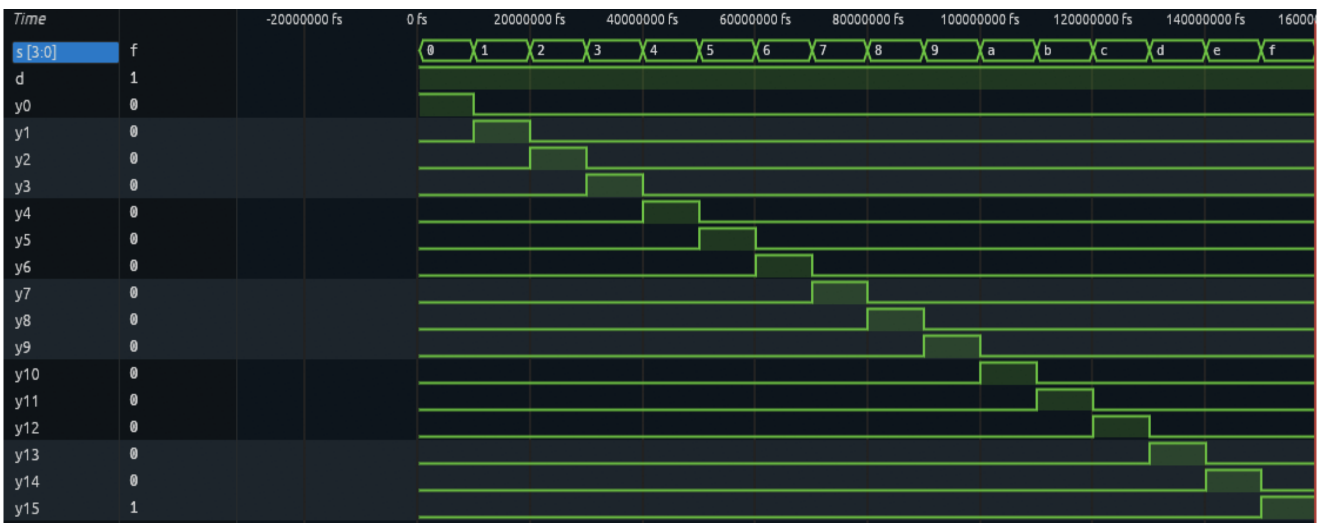
```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity tb_DMUX1to16 is
end tb_DMUX1to16;

architecture sim of tb_DMUX1to16 is

  signal D : std_logic := '0';
  signal S : std_logic_vector(3 downto 0);

  signal Y : std_logic_vector(15 downto 0);

  signal Y0  : std_logic;
  signal Y1  : std_logic;
  signal Y2  : std_logic;
  signal Y3  : std_logic;
  signal Y4  : std_logic;
  signal Y5  : std_logic;
  signal Y6  : std_logic;
  signal Y7  : std_logic;
  signal Y8  : std_logic;
  signal Y9  : std_logic;
  signal Y10 : std_logic;
  signal Y11 : std_logic;
  signal Y12 : std_logic;
  signal Y13 : std_logic;
  signal Y14 : std_logic;
  signal Y15 : std_logic;
```

```
begin

  uut : entity work.DMUX1to16
    port map
    (
      D => D,
      S => S,
      Y => Y
    );

  Y0  <= Y(0);
  Y1  <= Y(1);
  Y2  <= Y(2);
  Y3  <= Y(3);
  Y4  <= Y(4);
  Y5  <= Y(5);
  Y6  <= Y(6);
  Y7  <= Y(7);
  Y8  <= Y(8);
  Y9  <= Y(9);
  Y10 <= Y(10);
  Y11 <= Y(11);
  Y12 <= Y(12);
  Y13 <= Y(13);
  Y14 <= Y(14);
  Y15 <= Y(15);

  process
  begin

    D <= '1';

    S <= "0000";
```

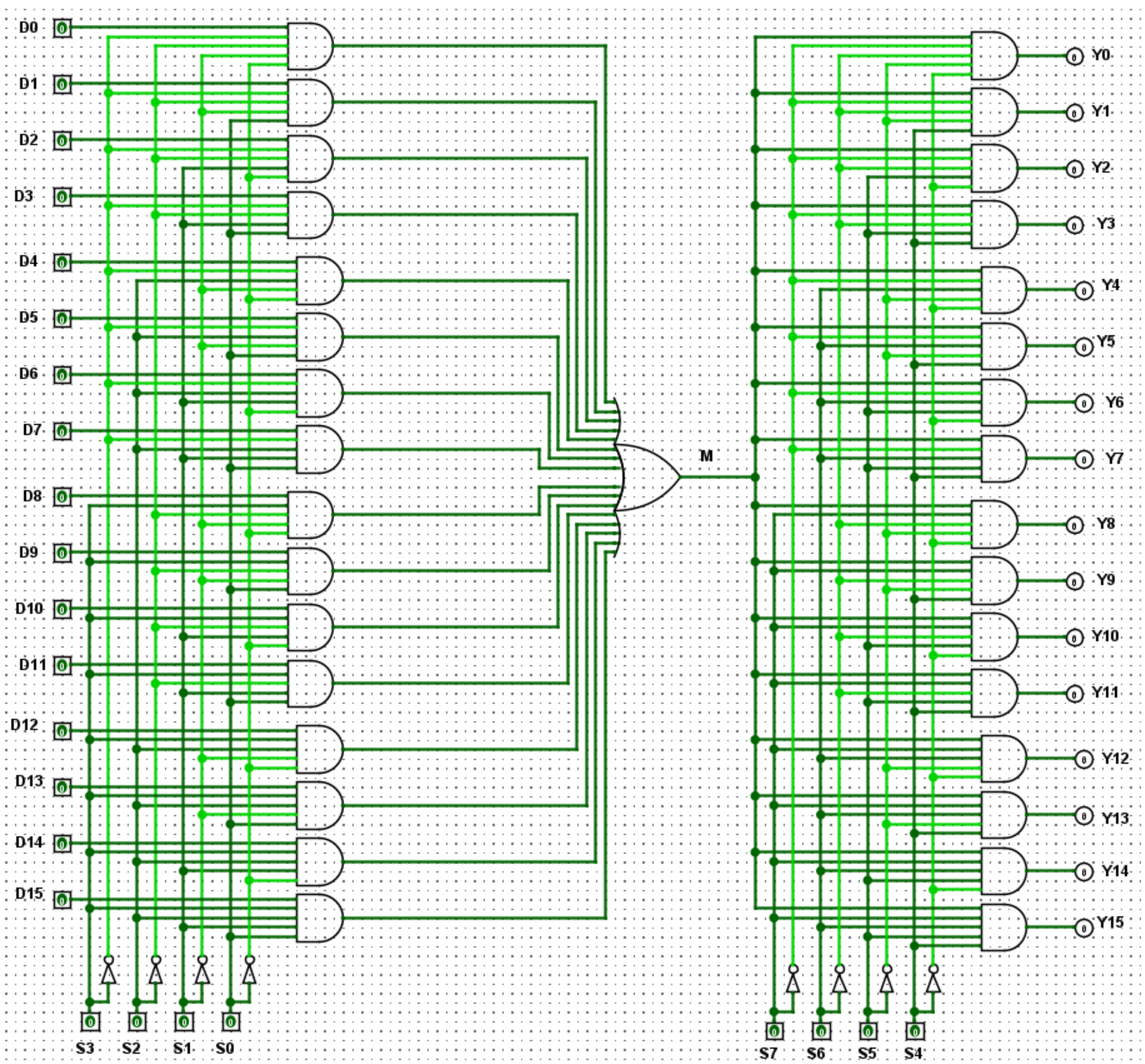
```
wait for 10 ns;
S <= "0001";
wait for 10 ns;
S <= "0010";
wait for 10 ns;
S <= "0011";
wait for 10 ns;

S <= "0100";
wait for 10 ns;
S <= "0101";
wait for 10 ns;
S <= "0110";
wait for 10 ns;
S <= "0111";
wait for 10 ns;

S <= "1000";
wait for 10 ns;
S <= "1001";
wait for 10 ns;
S <= "1010";
wait for 10 ns;
S <= "1011";
wait for 10 ns;

S <= "1100";
wait for 10 ns;
S <= "1101";
wait for 10 ns;
S <= "1110";
wait for 10 ns;
S <= "1111";
```

```
    wait for 10 ns;

    D <= '0';

    wait;

  end process;

end sim;
```

DMUX1-16-wave.vcd



mux_dmux16 curcuit



mux_dmux16.vhdl

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity mux_dmux16 is
  port (
    D : in std_logic_vector(15 downto 0);

    S_MUX  : in std_logic_vector(3 downto 0);
    S_DMUX : in std_logic_vector(3 downto 0);

    Y : out std_logic_vector(15 downto 0)
  );
end mux_dmux16;

architecture Structural of mux_dmux16 is

  signal M : std_logic;

begin

  ----------------------------------------------------------------
  -- MUX 16:1
  ----------------------------------------------------------------

  M <= (D(0) and not S_MUX(3) and not S_MUX(2) and not S_MUX(1) and
not S_MUX(0))
    or (D(1) and not S_MUX(3) and not S_MUX(2) and not S_MUX(1) and
S_MUX(0))
    or (D(2) and not S_MUX(3) and not S_MUX(2) and S_MUX(1) and not
S_MUX(0))
    or (D(3) and not S_MUX(3) and not S_MUX(2) and S_MUX(1) and
S_MUX(0))

    or (D(4) and not S_MUX(3) and S_MUX(2) and not S_MUX(1) and not
S_MUX(0))
```

```
    or (D(5) and not S_MUX(3) and S_MUX(2) and not S_MUX(1) and
S_MUX(0))
    or (D(6) and not S_MUX(3) and S_MUX(2) and S_MUX(1) and not
S_MUX(0))
    or (D(7) and not S_MUX(3) and S_MUX(2) and S_MUX(1) and S_MUX(0))

    or (D(8) and S_MUX(3) and not S_MUX(2) and not S_MUX(1) and not
S_MUX(0))
    or (D(9) and S_MUX(3) and not S_MUX(2) and not S_MUX(1) and
S_MUX(0))
    or (D(10) and S_MUX(3) and not S_MUX(2) and S_MUX(1) and not
S_MUX(0))
    or (D(11) and S_MUX(3) and not S_MUX(2) and S_MUX(1) and S_MUX(0))

    or (D(12) and S_MUX(3) and S_MUX(2) and not S_MUX(1) and not
S_MUX(0))
    or (D(13) and S_MUX(3) and S_MUX(2) and not S_MUX(1) and S_MUX(0))
    or (D(14) and S_MUX(3) and S_MUX(2) and S_MUX(1) and not S_MUX(0))
    or (D(15) and S_MUX(3) and S_MUX(2) and S_MUX(1) and S_MUX(0));

  ----------------------------------------------------------------
  -- DMUX 1:16
  ----------------------------------------------------------------

  Y(0) <= M and not S_DMUX(3) and not S_DMUX(2) and not S_DMUX(1) and
not S_DMUX(0);
  Y(1) <= M and not S_DMUX(3) and not S_DMUX(2) and not S_DMUX(1) and
S_DMUX(0);
  Y(2) <= M and not S_DMUX(3) and not S_DMUX(2) and S_DMUX(1) and not
S_DMUX(0);
  Y(3) <= M and not S_DMUX(3) and not S_DMUX(2) and S_DMUX(1) and
S_DMUX(0);

  Y(4) <= M and not S_DMUX(3) and S_DMUX(2) and not S_DMUX(1) and not
S_DMUX(0);
  Y(5) <= M and not S_DMUX(3) and S_DMUX(2) and not S_DMUX(1) and
S_DMUX(0);
  Y(6) <= M and not S_DMUX(3) and S_DMUX(2) and S_DMUX(1) and not
S_DMUX(0);
```

```
  Y(7) <= M and not S_DMUX(3) and S_DMUX(2) and S_DMUX(1) and
S_DMUX(0);

  Y(8)  <= M and S_DMUX(3) and not S_DMUX(2) and not S_DMUX(1) and not
S_DMUX(0);
  Y(9)  <= M and S_DMUX(3) and not S_DMUX(2) and not S_DMUX(1) and
S_DMUX(0);
  Y(10) <= M and S_DMUX(3) and not S_DMUX(2) and S_DMUX(1) and not
S_DMUX(0);
  Y(11) <= M and S_DMUX(3) and not S_DMUX(2) and S_DMUX(1) and
S_DMUX(0);

  Y(12) <= M and S_DMUX(3) and S_DMUX(2) and not S_DMUX(1) and not
S_DMUX(0);
  Y(13) <= M and S_DMUX(3) and S_DMUX(2) and not S_DMUX(1) and
S_DMUX(0);
  Y(14) <= M and S_DMUX(3) and S_DMUX(2) and S_DMUX(1) and not
S_DMUX(0);
  Y(15) <= M and S_DMUX(3) and S_DMUX(2) and S_DMUX(1) and S_DMUX(0);

end Structural;
```

tb_mux_dmux16.vhdl

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity tb_mux_dmux16 is
end tb_mux_dmux16;

architecture sim of tb_mux_dmux16 is

  signal D : std_logic_vector(15 downto 0);

  signal S_MUX  : std_logic_vector(3 downto 0);
  signal S_DMUX : std_logic_vector(3 downto 0);
```

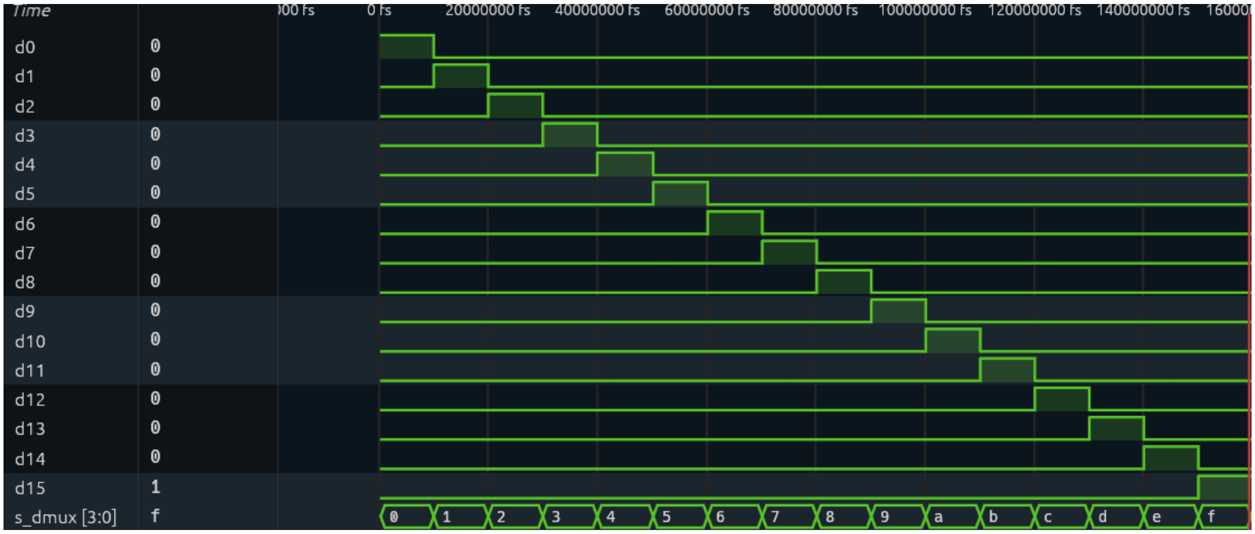
```
  signal Y : std_logic_vector(15 downto 0);

  -- แตก D ออกเป็นรายเส้น
  signal D0  : std_logic:= '0';
  signal D1  : std_logic:= '0';
  signal D2  : std_logic:= '0';
  signal D3  : std_logic:= '0';
  signal D4  : std_logic:= '0';
  signal D5  : std_logic:= '0';
  signal D6  : std_logic:= '0';
  signal D7  : std_logic:= '0';
  signal D8  : std_logic:= '0';
  signal D9  : std_logic:= '0';
  signal D10 : std_logic:= '0';
  signal D11 : std_logic:= '0';
  signal D12 : std_logic:= '0';
  signal D13 : std_logic:= '0';
  signal D14 : std_logic:= '0';
  signal D15 : std_logic:= '0';

  -- แตก Y ออกเป็นรายเส้น
  signal Y0  : std_logic;
  signal Y1  : std_logic;
  signal Y2  : std_logic;
  signal Y3  : std_logic;
  signal Y4  : std_logic;
  signal Y5  : std_logic;
  signal Y6  : std_logic;
  signal Y7  : std_logic;
  signal Y8  : std_logic;
  signal Y9  : std_logic;
  signal Y10 : std_logic;
  signal Y11 : std_logic;
  signal Y12 : std_logic;
  signal Y13 : std_logic;
  signal Y14 : std_logic;
  signal Y15 : std_logic;

begin
```

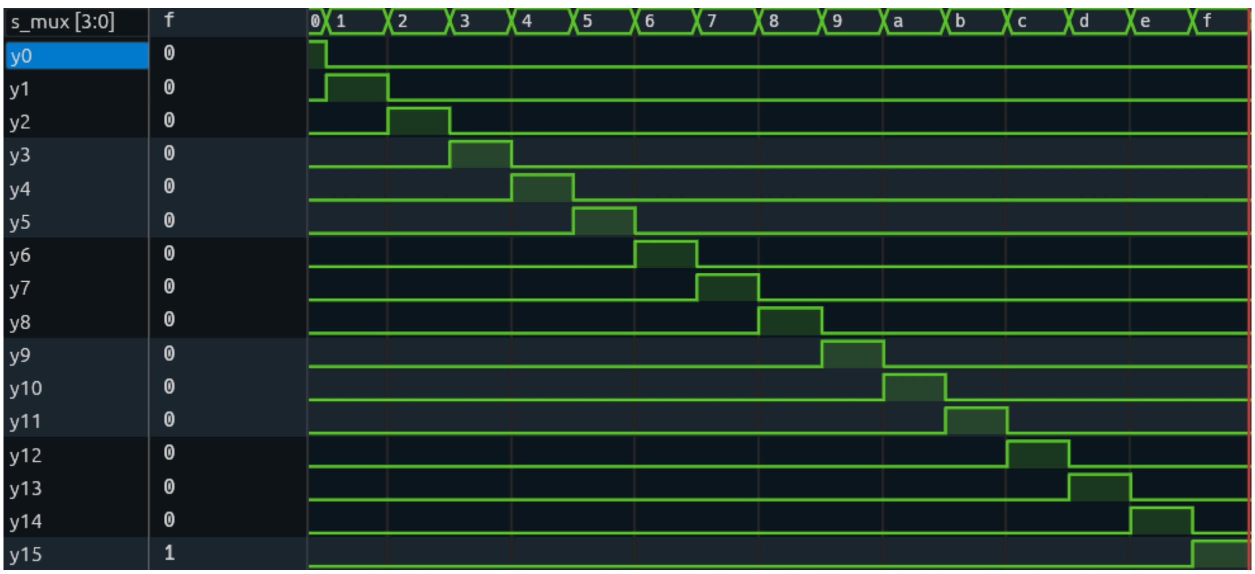
```
----------------------------------------------------------------
-- รวม D0-D15 เป็น Bus
----------------------------------------------------------------

D <= D15 & D14 & D13 & D12 &
  D11 & D10 & D9 & D8 &
  D7 & D6 & D5 & D4 &
  D3 & D2 & D1 & D0;

----------------------------------------------------------------
-- แยก Y
----------------------------------------------------------------

Y0  <= Y(0);
Y1  <= Y(1);
Y2  <= Y(2);
Y3  <= Y(3);
Y4  <= Y(4);
Y5  <= Y(5);
Y6  <= Y(6);
Y7  <= Y(7);
Y8  <= Y(8);
Y9  <= Y(9);
Y10 <= Y(10);
Y11 <= Y(11);
Y12 <= Y(12);
Y13 <= Y(13);
Y14 <= Y(14);
Y15 <= Y(15);

----------------------------------------------------------------
-- DUT
----------------------------------------------------------------

uut : entity work.mux_dmux16
  port map
  (
    D      => D,
```

```
    S_MUX  => S_MUX,
    S_DMUX => S_DMUX,
    Y      => Y
  );

-----------------------------------------------------------------
-- Truth Table Test
-----------------------------------------------------------------

process
begin

  D0     <= '1';
  S_MUX  <= "0000";
  S_DMUX <= "0000";
  wait for 10 ns;
  D0 <= '0';

  D1     <= '1';
  S_MUX  <= "0001";
  S_DMUX <= "0001";
  wait for 10 ns;
  D1 <= '0';

  D2     <= '1';
  S_MUX  <= "0010";
  S_DMUX <= "0010";
  wait for 10 ns;
  D2 <= '0';

  D3     <= '1';
  S_MUX  <= "0011";
  S_DMUX <= "0011";
  wait for 10 ns;
  D3 <= '0';

  D4     <= '1';
  S_MUX  <= "0100";
  S_DMUX <= "0100";
```

```
wait for 10 ns;
D4 <= '0';

D5     <= '1';
S_MUX  <= "0101";
S_DMUX <= "0101";
wait for 10 ns;
D5 <= '0';

D6     <= '1';
S_MUX  <= "0110";
S_DMUX <= "0110";
wait for 10 ns;
D6 <= '0';

D7     <= '1';
S_MUX  <= "0111";
S_DMUX <= "0111";
wait for 10 ns;
D7 <= '0';

D8     <= '1';
S_MUX  <= "1000";
S_DMUX <= "1000";
wait for 10 ns;
D8 <= '0';

D9     <= '1';
S_MUX  <= "1001";
S_DMUX <= "1001";
wait for 10 ns;
D9 <= '0';

D10    <= '1';
S_MUX  <= "1010";
S_DMUX <= "1010";
wait for 10 ns;
D10 <= '0';
```

```
    D11    <= '1';
    S_MUX  <= "1011";
    S_DMUX <= "1011";
    wait for 10 ns;
    D11 <= '0';

    D12    <= '1';
    S_MUX  <= "1100";
    S_DMUX <= "1100";
    wait for 10 ns;
    D12 <= '0';

    D13    <= '1';
    S_MUX  <= "1101";
    S_DMUX <= "1101";
    wait for 10 ns;
    D13 <= '0';

    D14    <= '1';
    S_MUX  <= "1110";
    S_DMUX <= "1110";
    wait for 10 ns;
    D14 <= '0';

    D15    <= '1';
    S_MUX  <= "1111";
    S_DMUX <= "1111";
    wait for 10 ns;
    D15 <= '0';

    wait;

  end process;

end sim;
```


Time
0 fs
20000000 fs
40000000 fs
60000000 fs
80000000 fs
100000000 fs
120000000 fs
140000000 fs
d0 0
d1 0
d2 0
d3 0
d4 0
d5 0
d6 0
d7 0
d8 0
d9 0
d10 0
d11 0
d12 0
d13 0
d14 0
d15 1
s_dmux [3:0] f
0 1 2 3 4 5 6 7 8 9 a b c d e f


s_mux [3:0] f
0 1 2 3 4 5 6 7 8 9 a b c d e f
y0 0
y1 0
y2 0
y3 0
y4 0
y5 0
y6 0
y7 0
y8 0
y9 0
y10 0
y11 0
y12 0
y13 0
y14 0
y15 1